Bivariate Data: PMCC Hypothesis Testing

Questions 1–15

Question 1. Tables of critical values are on page 356 Describe what is meant by the significance level.

Question 2. In each of the following cases, r is the sample correlation coefficient. Determine whether you reject H_0 or fail to reject H_0

a) $H_0 : \rho = 0$, $H_1 : \rho > 0$
The test statistic is $r = 0.638$ and the p-value is 0.0444, where $n = 8$ and the significance level is 5%

b) $H_0 : \rho = 0$, $H_1 : \rho < 0$
The test statistic is $r = -0.456$ and the p-value is 0.0681, where $n = 12$ and the significance level is 5%

Question 3. In each of the following cases, r is the sample correlation coefficient. Determine whether you reject H_0 or fail to reject H_0

a) $H_0 : \rho = 0$, $H_1 : \rho > 0$
$r = 0.487$
Critical value for $n = 14$ for a one-tailed test at a 5% significance level is 0.4575

b) $H_0 : \rho = 0$, $H_1 : \rho > 0$
$r = 0.628$
Critical value for $n = 7$ for a one-tailed test at a 5% significance level is 0.6694

c) $H_0 : \rho = 0$, $H_1 : \rho > 0$
$r = 0.593$
Critical value for $n = 20$ for a one-tailed test at a 1% significance level is 0.5155

d) $H_0 : \rho = 0$, $H_1 : \rho > 0$
$r = 0.213$
Critical value for $n = 60$ for a one-tailed test at a 10% significance level is 0.1678

Question 4. In each of the following cases, r is the sample correlation coefficient. Determine whether you reject H_0 or fail to reject H_0

a) $H_0 : \rho = 0$, $H_1 : \rho < 0$
$r = -0.453$
Critical value for $n = 15$ for a one-tailed test at a 5% significance level is 0.4409

b) $H_0 : \rho = 0$, $H_1 : \rho < 0$
$r = -0.829$
Critical value for $n = 7$ for a one-tailed test at a 5% significance level is 0.6694

c) $H_0 : \rho = 0$, $H_1 : \rho < 0$
$r = -0.306$
Critical value for $n = 25$ for a one-tailed test at a 1% significance level is 0.4622

d) $H_0 : \rho = 0$, $H_1 : \rho < 0$
$r = -0.197$
Critical value for $n = 40$ for a one-tailed test at a 10% significance level is 0.2070

Question 5. State the critical region for the following test:
$H_0 : \rho = 0$, $H_1 : \rho > 0$
Critical value for $n = 19$ for a one-tailed test at a 0.5% significance level is 0.5751

Question 6. State the critical region for the following test:
$H_0 : \rho = 0$, $H_1 : \rho < 0$
Critical value for $n = 16$ for a one-tailed test at a 1% significance level is 0.5742

Question 7. In each of the following cases, r is the sample correlation coefficient. Determine whether you reject H_0 or fail to reject H_0

a) $H_0 : \rho = 0$, $H_1 : \rho \neq 0$
$r = 0.658$
Critical value for $n = 9$ for a two-tailed test at a 5% significance level is 0.6664

b) $H_0 : \rho = 0$, $H_1 : \rho \neq 0$
$r = -0.567$
Critical value for $n = 11$ for a two-tailed test at a 5% significance level is 0.6021

c) $H_0 : \rho = 0$, $H_1 : \rho \neq 0$
$r = 0.621$
Critical value for $n = 16$ for a two-tailed test at a 1% significance level is 0.6226

d) $H_0 : \rho = 0$, $H_1 : \rho \neq 0$
$r = -0.248$
Critical value for $n = 50$ for a two-tailed test at a 10% significance level is 0.2353

Question 8. State the critical region for the following test:
$H_0 : \rho = 0$, $H_1 : \rho \neq 0$
Critical value for $n = 7$ for a two-tailed test at a 2.5% significance level is 0.7545

Question 9. State the critical region for the following test:
$H_0 : \rho = 0$, $H_1 : \rho \neq 0$
Critical value for $n = 80$ for a two-tailed test at a 5% significance level is 0.2199

Question 10. Hannah believes there is a positive correlation between the amount of sugar consumed daily by adults and their blood glucose levels. She collects a random sample of 28 adults and finds the sample's correlation coefficient to be 0.382. Carry out a hypothesis test at the 5% significance level to investigate whether Hannah's belief is supported.

Question 11. Kyle is a social scientist who wants to explore if there is a negative correlation between the number of hours spent on social media each day and self-reported feelings of loneliness in teenagers. He collects a random sample of 70 teenagers and finds the sample's correlation coefficient to be -0.297. Carry out a hypothesis test at the 0.5% significance level to investigate whether there is evidence to support a negative correlation.

Question 12. Hamza is a biologist who is investigating whether there is a negative correlation between the water temperature they grow up in and their growth rate. Hamza wants to conduct a hypothesis test at the 1% significance level. He collects a random sample of 40 of a certain type of fish and finds the sample's correlation coefficient to be -0.364. Hamza's hypothesis test is shown below.

Let ρ be the sample correlation coefficient between the water temperature and the growth rate of the fish.

$H_0 : \rho = 0$

$H_1 : \rho < 0$

$r = -0.364$

The critical value for $n = 40$ for a one-tailed test at a 1% significance level is 0.3665

$-0.364 < 0.3665$ *so we reject* H_0

There is sufficient evidence to suggest that there is a negative correlation between the water temperature and the growth rate of the fish.

a) Identify the first two errors made by Hamza in his hypothesis test.

b) Correct his test.

Question 13. Aida is an environmental scientist who wants to determine if there is some correlation between the levels of air pollution and numbers of adults with respiratory illnesses. She collects data from 50 randomly selected neighbourhoods and finds the sample correlation coefficient to be 0.209. Carry out a hypothesis test at the 5% significance level to investigate whether there is any evidence to support there being some correlation.

Question 14. Kaylee is a nutritionist who wants to determine if there is any correlation between the daily intake of fruits and vegetables in adults and their body mass index (BMI). She collects data from 100 randomly selected adults and finds the sample correlation coefficient to be -0.301. Carry out a hypothesis test at the 1% significance level to investigate whether there is any evidence to suggest a correlation.

Question 15. Jo is a psychologist who is investigating whether there is any correlation between the number of hours of sleep college students get each night and their stress levels, as measured by a standardised test. Jo wants to conduct a hypothesis test at the 5% significance level. She collects data from 28 randomly selected college students and finds the sample correlation coefficient to be 0.482. Jo's hypothesis test is shown below.

Let ρ be the population correlation coefficient between the number of hours of sleep college students get each night and their stress levels.

$H_0 : \rho = 0$

$H_1 : \rho \neq 0$

$r = 0.482$

The critical value for $n = 28$ for a two-tailed test at a 5% significance level is 0.3172

$0.3172 < 0.482$ *so we fail to reject* H_0

There is insufficient evidence to suggest that there is some correlation between the number of hours of sleep college students get each night and their stress levels.

a) Identify the first two errors made by Jo in her hypothesis test.

b) Correct her test.